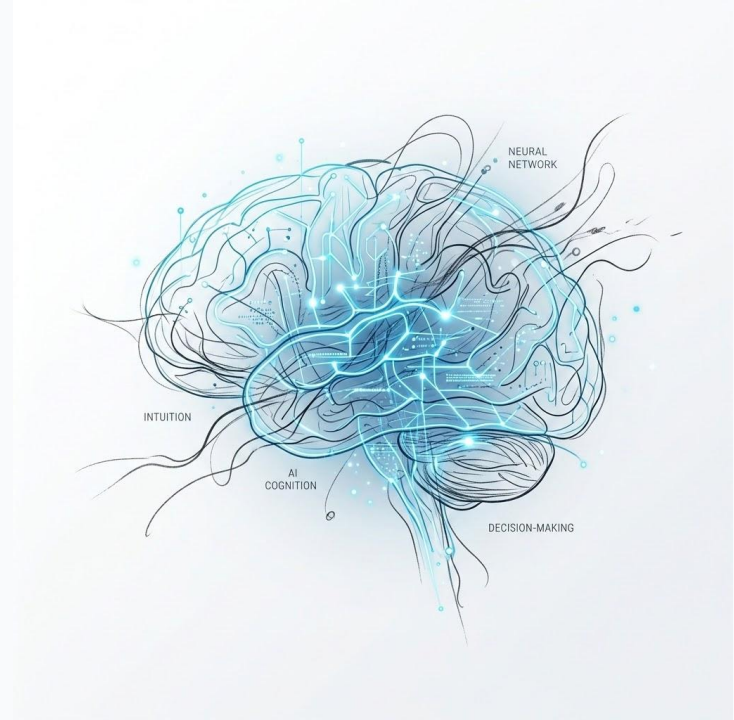


How Does a Computer Actually Work?

SYS 478. Fall 2026. Prof. Sarah Van Wart



Outline

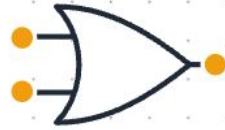
1. Hardware (electricity and silicon + Von Neumann Architecture)
2. Materials
3. Software (using actual language with can convert to instructions)

Activity 1: Build a System Demo

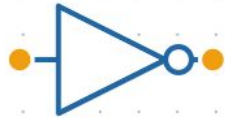
Logic Gates



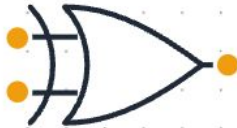
AND: Output is true if **both** inputs are true
(2 transistors in series)



OR: Output is true if **at least one** input is true
(2 transistors in parallel)



NOT: **Inverts** the input
(1 transistor)



XOR: Output is true if **either input** is true but not both
(3 transistors: NOT + OR)

Discussion Questions

1. **Representation:** What facts about the world became inputs? What was left out?
2. **Procedure:** Where is the rule that determines what happens?
3. **Values:** What outcome is the system designed to prioritize – saving water, keeping plants alive, convenience?
4. **Accountability:** If the sprinkler behaves badly, can we reconstruct why?
5. **Limits:** When would you not want this system to act automatically?

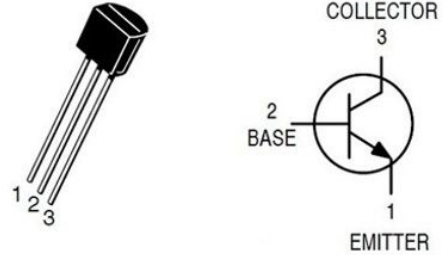
Transistors & Logic Gates

Transistors

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power.

Structure of a Transistor:

- **Emitter:** Releases electrons.
- **Base:** Controls electron flow.
- **Collector:** Collects electrons
 - By applying a voltage at the collector, it "pulls" electrons through the transistor, creating a larger current that flows from the collector to the emitter.



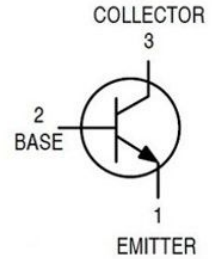
Transistors

Applications:

- Switching: Controls on/off states in circuits.
- Amplification: Boosts audio and radio signals.

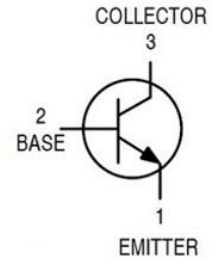
Importance:

- Enables miniaturization of devices, making powerful and compact electronics possible.
- Transistors are the building blocks of logic gates, which are the fundamental components in digital circuits that perform logical operations (like AND, OR, NOT)



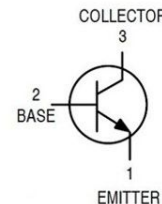
Transistor fun facts

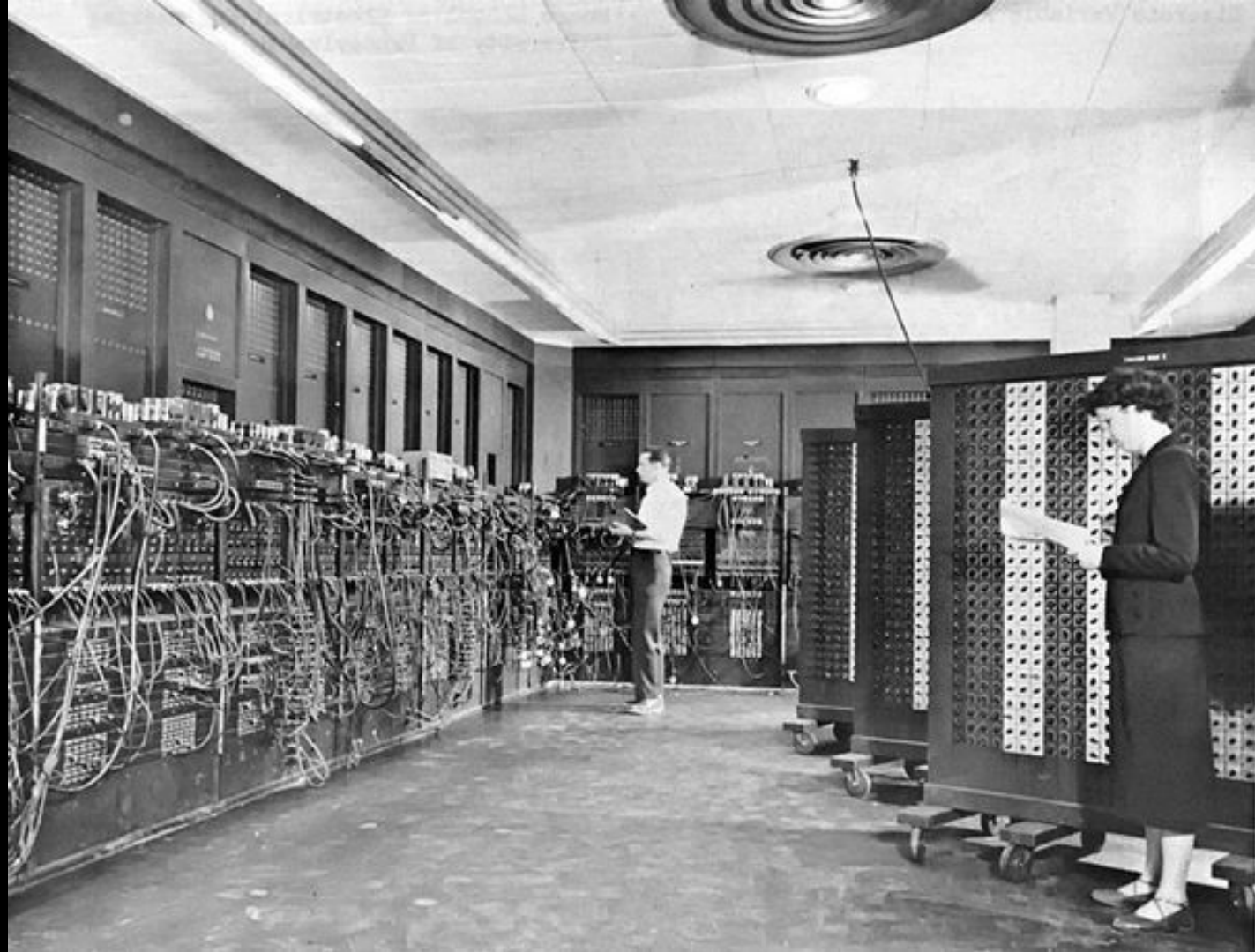
- Transistors are everywhere inside a computer – wherever the machine needs to compute, control, or electronically store information.
- A modern processor has billions to tens of billions of them.
- Transistors are made of silicon (sand), a semiconductor. Manufacturers alter tiny regions of silicon by adding controlled impurities – called doping – so that they can control when and where current flows



Transistor fun facts

- Where are they? CPU, GPU, RAM, flash storage. Basically, almost every digital electronic device
- A modern processor has **billions to tens of billions** of them.
- Transistors are **made of silicon**, a semiconductor. Manufacturers alter tiny regions of silicon by adding controlled impurities – called doping – so that they can control when and where current flows
- **They're small.** A human hair is roughly 50,000–100,000 nanometers wide. Key structures inside advanced transistors can be only a few to a few tens of nanometers across.





Short Timeline

- **1940s:** stored-program computer architecture emerges
- **Von Neumann model:** instructions and data stored in memory
- **Modern computers:** still broadly organized around processor, memory, storage, and input/output

The stored-program idea: Von Neumann

Both data and instructions can be stored in memory. The processor retrieves and executes those instructions

BIG IDEA: **The same physical machine can do many different things** without having to manually rewire everything (because instructions can also be represented by electrical signals – instead of by wires)

Fetch → Decode → Execute

The processor repeatedly:

- Fetches the next instruction
- Decodes what the instruction means
- Executes it
- Moves to the next instruction

Possible example:

- LOAD 5
- ADD 3
- STORE result
- DISPLAY result

Act it out.

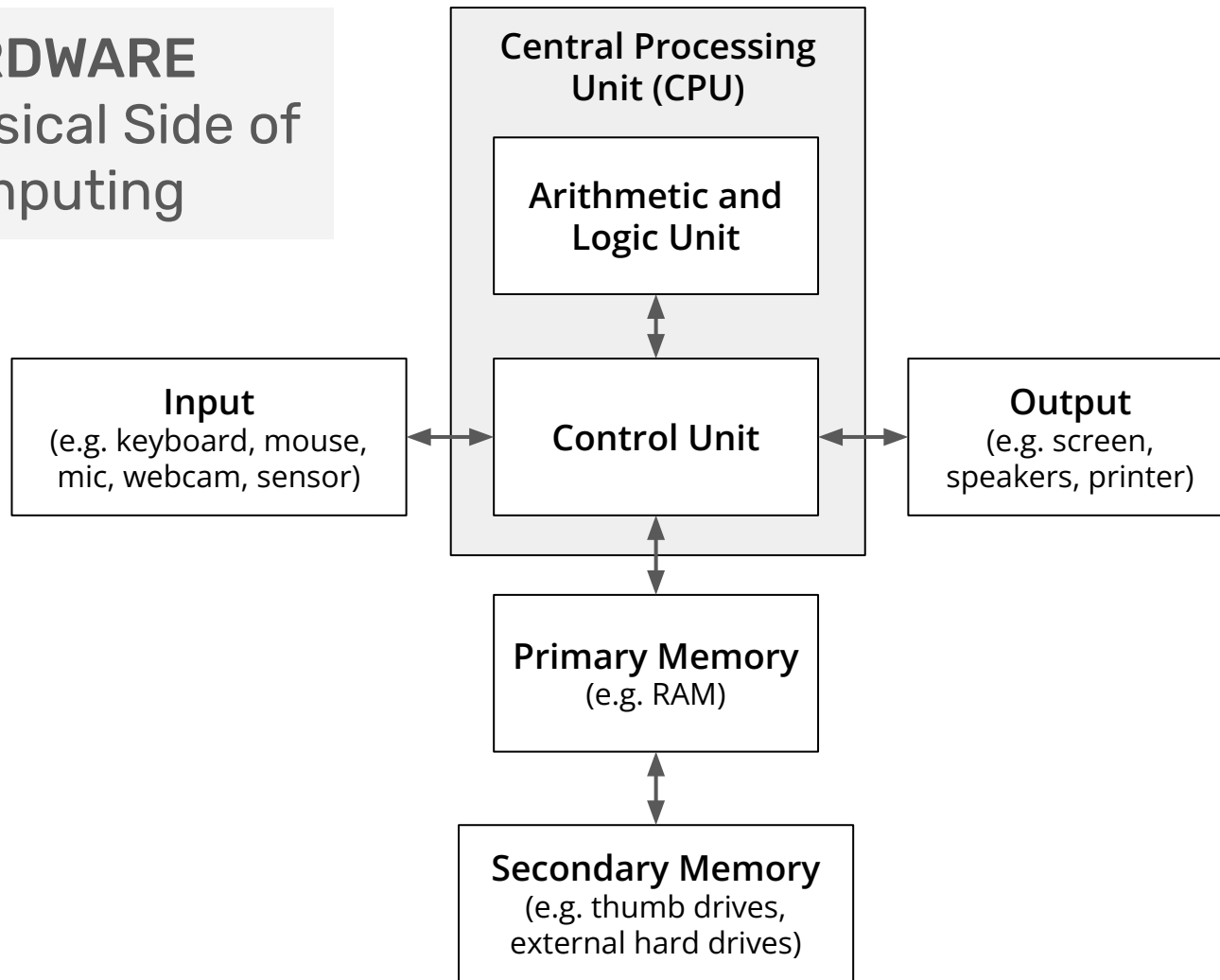
- One student = memory
- One student = processor
- One student = input/output

3 Big Components of a Computer

- **Memory:** Stores data and instructions
- **CPU** (Central Processing Unit): Executes instructions
- **Input/Output:** Interfaces for data to enter and exit the system

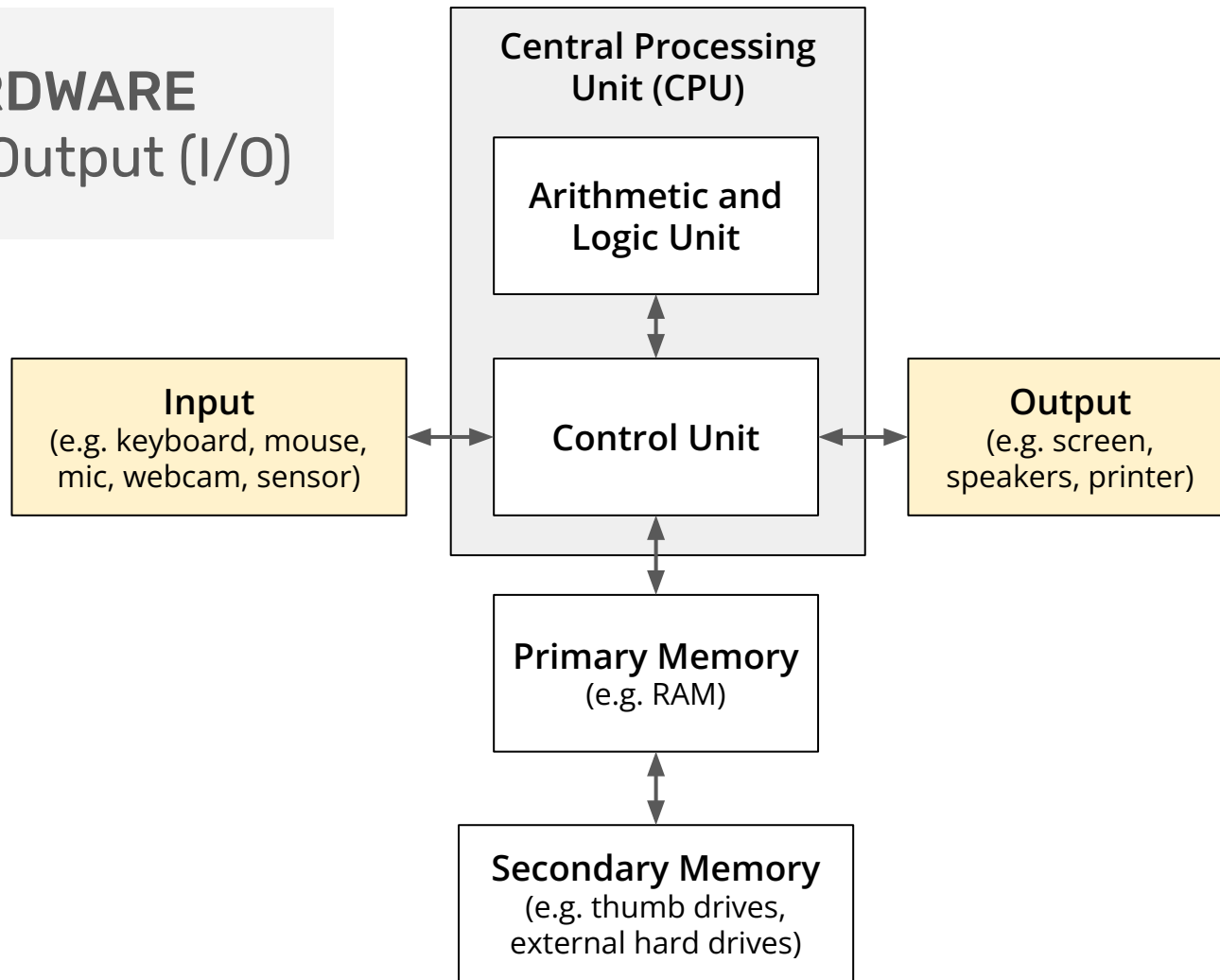
HARDWARE

The Physical Side of Computing



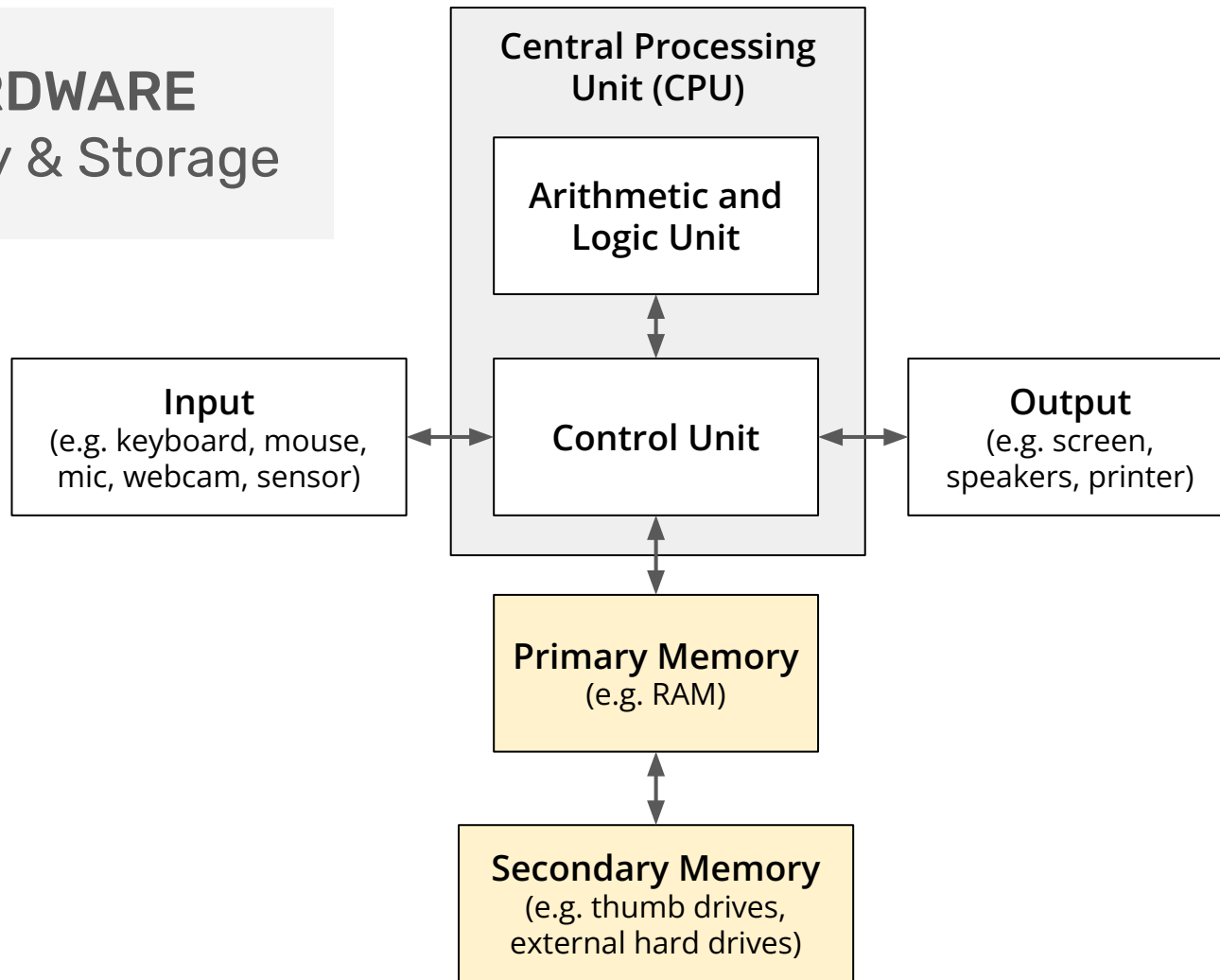
HARDWARE

Input / Output (I/O)

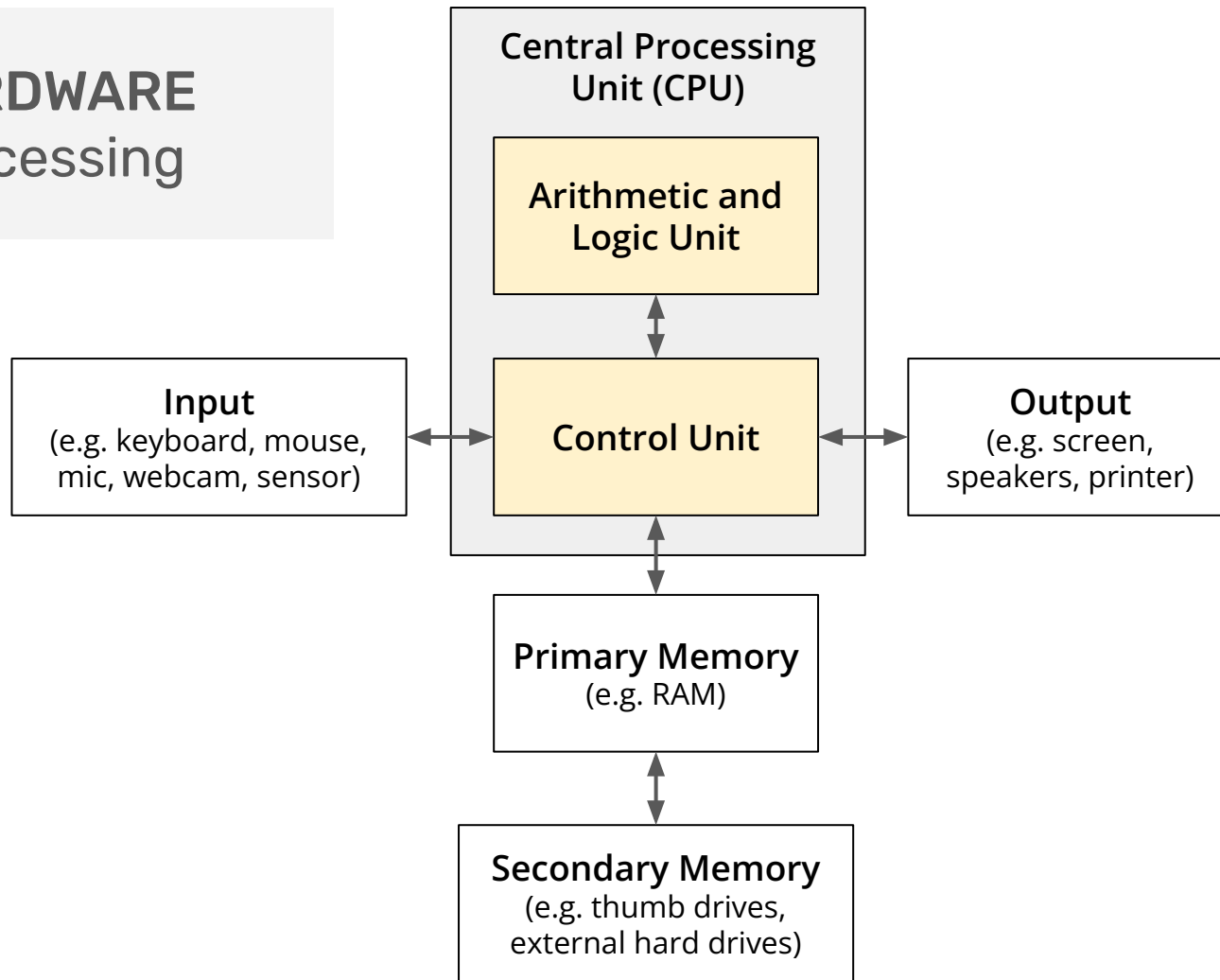


HARDWARE

Memory & Storage



HARDWARE Processing



Scale Changes the Stakes

From One Computer to Billions of Connected Computers

How many computers do you touch each day?

Today, many of the things we do on a “computer” are not happening on just one machine.

- Your phone or laptop often sends data across a network to other computers somewhere else.
- Those remote computers: store data, run software, process requests, recommend content, stream video, serve ads, train and run AI models

The Internet Runs on Physical Machines

Service

- Streaming Netflix
- Scrolling TikTok or Instagram
- Watching YouTube
- Searching Google
- Saving files to the cloud
- Playing online games
- Using generative AI

Physical Infrastructure

- servers
- processors
- RAM
- storage
- networking equipment
- electricity
- cooling
- buildings
- fiber-optic cables

Electricity

- Data centers use roughly **1.5% of global electricity demand**.
- By 2030, around **3% of global demand**.
- AI-focused data centers are one of the major drivers of that growth.
- Communities where data centers are physically located bear most of the immediate costs



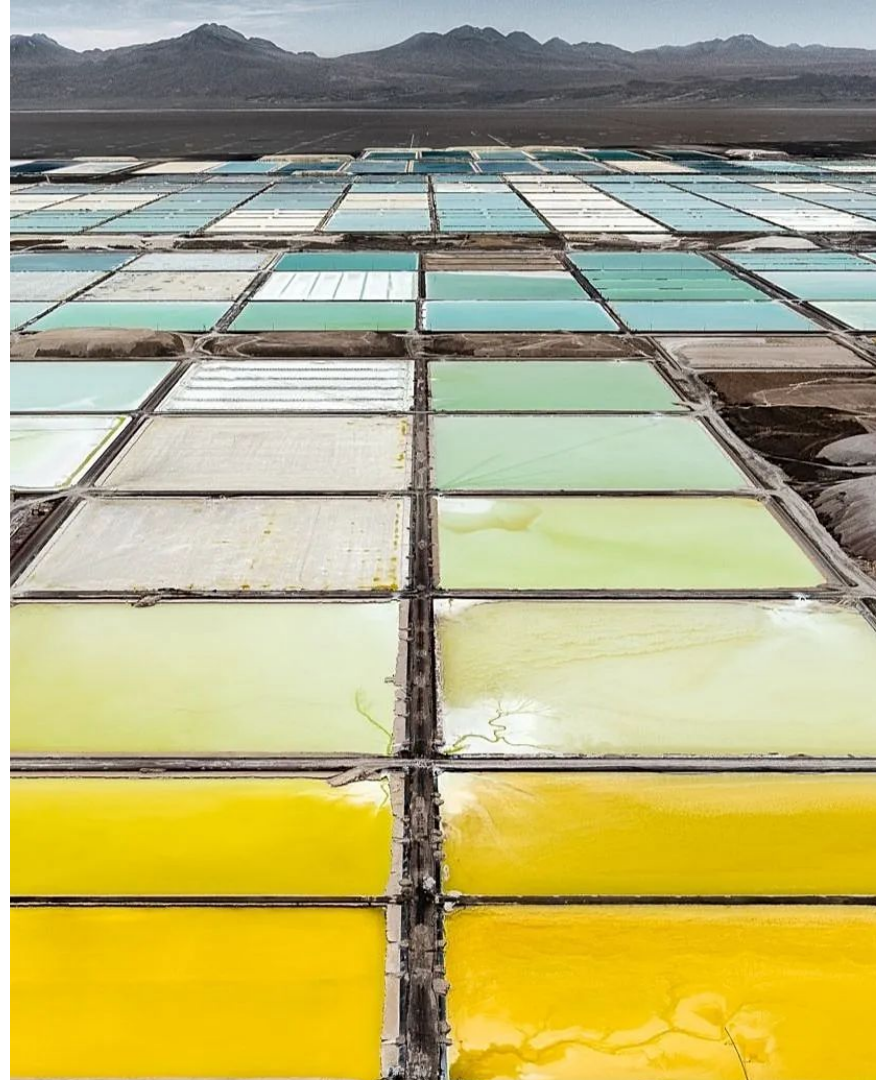
Water

- Electrical energy used by chips ultimately produces heat that data centers have to remove that heat continuously.
- Water demand varies substantially by location, climate, cooling design, and electricity source.
- Water is also needed for manufacturing (e.g., semiconductor fabrication requires extremely pure water).



Materials

- **Silicon:** semiconductor material used to make transistors.
- **Copper:** wiring, circuit boards, electrical distribution, transformers, and grid infrastructure.
- **Lithium, nickel, cobalt, graphite:** batteries and energy-storage systems.
- **Gallium and germanium:** specialized semiconductor applications.
- **Mining and refining:** supply-chain and geopolitical dependencies.



Waste

- Servers, GPUs, batteries, and networking equipment eventually become e-waste.
- Rapid hardware upgrades can shorten equipment lifespans.
- E-waste contains both valuable materials and hazardous substances.
- Recycling is difficult because many materials are tightly integrated.



Which uses of computers are more / less valid?

- Is it useful to audit and track which uses of materials, energy, water are used across the entire lifecycle?
 - Netflix v. Youtube v. ChatGPT v. Claude?
- How do we decide which uses are legitimate uses of energy, water, minerals, and which are not?
- Where do environmental considerations around computation fall within the larger arc of environmental justice?